Unofficial RPI Poster Theme for Beamer

Inwon Kang [1] GPT 5.5 [2]

[1]Rensselaer Polytechnic Institute [2]OpenAI

Layout commands

These helpers make poster geometry explicit.

- `pframe`: one Beamer frame plus left/right gutters.
- `pcolumn{width}`: a top-aligned minipage of exact width.
- `prow`: puts columns on one row; use it for nested columns.
- `pblock[style]{title}{height}`: block with a target height.

Use `alert` or `example` as optional block styles. If content is taller than the target, the block grows instead of overprinting the next block.

Usual skeleton

`pframe` $\to$ `pcolumn` $\to$ `pblock`

Nested layout: `prow` $\to$ `pcolumn` $\to$ `pblock`.

Width recipe

Pick the number of columns first, then solve for widths.

- Top level: $(N+1)s + Nw =$ `\textwidth`.
- Nested row: $\sum w_i + (N-1)s =$ `parent width`.
- Here: $s =$ `\pcolumnsepwidth` and $w =$ `\colwidth`.

Why not Beamer columns?

Beamer's `columns` is made for slides, not grid-filling posters.

- It adds internal spacing, so the visible width is not just the number you typed.
- Column height follows content; it does not reserve an exact block height.
- Nested layouts are harder to make flush with the outer grid.

The custom helpers use plain minipages and fixed gutters, so the math stays predictable.

Height recipe

Give every block a target height and let the last block fill the remainder.

$$\sum h_i + (M-1)\,\texttt{\textbackslash blocksepheight} = \texttt{\textbackslash textheight}$$

For a column with blocks of 24cm, `12.5cm`, and one remaining block:

```
\textheight - 24cm - 12.5cm - 2\blocksepheight
```

Use	Pattern
Same-height stack	choose fixed cm values
Fill leftover space	subtract used heights from `\textheight`
Two-column span	`2\colwidth + \pcolumnsepwidth`
Too much content	block grows; shorten text or increase height

Keep these formulas near the length definitions in the preamble.

Changing type size

Global sizes live in `beamerthemerpi.sty`.

- `block title`: usually `\Large` and bold.
- `block body`: usually `\large`.
- `headline title`: usually `\Huge`.

Local override: `{\Large ...}`. Exact size: `\fontsize{34}{40}\selectfont`.

Changing separator lengths

Two lengths control most whitespace.

- `\pcolumnsepwidth`: horizontal gutter between columns.
- `\blocksepheight`: vertical gap between stacked blocks.

Set them in the preamble:

```
\setlength{\pcolumnsepwidth}{0.025\textwidth}
\setlength{\blocksepheight}{4ex}
```

Mini layout checklist

1. Choose poster size in `beamerposter`.
2. Choose `\pcolumnsepwidth`.
3. Solve for `\colwidth`.
4. Set span widths, e.g. `\twocolwidth`.
5. Pick block heights to sum to `\textheight`.
6. Build, then trim text or adjust heights.

Rule of thumb

Make the grid first, then write to fit it. Clean posters have shared gutters, aligned bottoms, and short text.

References

[1] Claude E. Shannon.
A mathematical theory of communication.
Bell System Technical Journal, 27(3):379–423, 1948.